

**Windows Fundamentals**

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Operating System Structure

In Windows operating systems, the root directory is `<drive_letter>:\` (commonly C drive). The root directory (also known as the boot partition) is where the operating system is installed. Other physical and virtual drives are assigned other letters, for example, Data (E:). The directory structure of the boot partition is as follows:

Directory	Function
Perflogs	Can hold Windows performance logs but is empty by default.
Program Files	On 32-bit systems, all 16-bit and 32-bit programs are installed here. On 64-bit systems, only 64-bit programs are installed here.
Program Files (x86)	32-bit and 16-bit programs are installed here on 64-bit editions of Windows.
ProgramData	This is a hidden folder that contains data that is essential for certain installed programs to run. This data is accessible by the program no matter what user is running it.

Directory	Function
Users	This folder contains user profiles for each user that logs onto the system and contains the two folders Public and Default.
Default	This is the default user profile template for all created users. Whenever a new user is added to the system, their profile is based on the Default profile.
Public	This folder is intended for computer users to share files and is accessible to all users by default. This folder is shared over the network by default but requires a valid network account to access.
AppData	Per user application data and settings are stored in a hidden user subfolder (i.e., cliff.moore\AppData). Each of these folders contains three subfolders. The Roaming folder contains machine-independent data that should follow the user's profile, such as custom dictionaries. The Local folder is specific to the computer itself and is never synchronized across the network. LocalLow is similar to the Local folder, but it has a lower data integrity level. Therefore it can be used, for example, by a web browser set to protected or safe mode.
Windows	The majority of the files required for the Windows operating system are contained here.
System, System32, SysWOW64	Contains all DLLs required for the core features of Windows and the Windows API. The operating system searches these folders any time a program asks to load a DLL without specifying an absolute path.
WinSxS	The Windows Component Store contains a copy of all Windows components, updates, and service packs.

Exploring Directories Using Command Line

We can explore the file system using the **dir** command.

cmd-session

```
C:\htb> dir c:\ /a
```

```
Volume in drive C has no label.
```

```
Volume Serial Number is F416-77BE
```

```
Directory of c:\
```

08/16/2020	10:33 AM	<DIR>	\$Recycle.Bin
06/25/2020	06:25 PM	<DIR>	\$WinREAgent
07/02/2020	12:55 PM		1,024 AMTAG.BIN
06/25/2020	03:38 PM	<JUNCTION>	Documents and Settings [C:\Users]
08/13/2020	06:03 PM		8,192 DumpStack.log
08/17/2020	12:11 PM		8,192 DumpStack.log.tmp
08/27/2020	10:42 AM		37,752,373,248 hiberfil.sys
08/17/2020	12:11 PM		13,421,772,800 pagefile.sys
12/07/2019	05:14 AM	<DIR>	PerfLogs
08/24/2020	10:38 AM	<DIR>	Program Files
07/09/2020	06:08 PM	<DIR>	Program Files (x86)
08/24/2020	10:41 AM	<DIR>	ProgramData
06/25/2020	03:38 PM	<DIR>	Recovery
06/25/2020	03:57 PM		2,918 RHDSetup.log
08/17/2020	12:11 PM		16,777,216 swapfile.sys
08/26/2020	02:51 PM	<DIR>	System Volume Information
08/16/2020	10:33 AM	<DIR>	Users
08/17/2020	11:38 PM	<DIR>	Windows

```
7 File(s) 51,190,943,590 bytes
13 Dir(s) 261,310,697,472 bytes free
```

The **tree** utility is useful for graphically displaying the directory structure of a path or disk.

cmd-session

```
C:\htb> tree "c:\Program Files (x86)\VMware"
```

```
Folder PATH listing
```

```
Volume serial number is F416-77BE
```

```
C:\PROGRAM FILES (X86)\VMWARE
```

```
├──VMware VIX
│   ├──doc
│   │   ├──errors
│   │   ├──features
│   │   ├──lang
│   │   │   └──c
│   │   │       └──functions
│   │   └──types
│   ├──samples
│   └──Workstation-15.0.0
│       ├──32bit
│       └──64bit
└──VMware Workstation
    ├──env
    ├──hostd
    │   ├──coreLocale
    │   │   └──en
    │   ├──docroot
    │   └──client
```

```

├── sdk
├── extensions
│   ├── hostdiag
│   │   └── locale
│   │       └── en
│   └── vimLocale
│       └── en
├── ico
├── messages
│   ├── ja
│   └── zh_CN
├── OVFTool
│   ├── env
│   │   └── en
│   └── schemas
│       ├── DMTF
│       └── vmware
├── Resources
├── tools-upgraders
└── x64

```

```
tree c:\ /f | more
```

Connect to HTB

Target(s)

Time left: 108 min(s)



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